

Orthic Solonchaks

Description:

Orthic Solonchaks are strongly saline soils commonly found in arid and semi-arid regions where evaporation exceeds precipitation. They are characterized by a high concentration of soluble salts (mainly chlorides and sulfates of sodium, calcium, and magnesium) within the upper 50 cm of the soil profile, giving the surface a white to light gray crust in dry periods.

These soils typically form on fine-textured parent materials such as clay-rich alluvium, lacustrine deposits, or silty plains with poor natural drainage.

The term “Orthic” indicates that the saline horizon is relatively uniform and lacks significant horizon development or carbonate accumulation.

Drainage:

- **Wet season:** Drainage is often impeded by shallow clay layers or subsurface salt accumulation. Waterlogging may occur after heavy rains.
- **Dry season:** Soils dry quickly, leaving a saline crust at the surface that can hinder seed germination and emergence. Evaporation concentrates salts near the surface, increasing soil salinity.

Landscape:

Found on low-lying plains, lake shores, closed depressions, and old floodplains in semi-arid and arid regions. These soils occur extensively in Northern and Eastern Kenya (Garissa, Wajir, Tana River), the Rift Valley lowlands, and saline depressions in the coastal lowlands. Annual rainfall is typically 200–600 mm.

Depth:

Usually shallow to moderately deep (30–100 cm). Rooting depth is often restricted by high salt concentrations or a shallow clay/salt layer, limiting the growth of sensitive crops.

Workability:

- **When wet:** Sticky and plastic due to high clay content; difficult to till.
- **When dry:** Hard, crusted, and saline; surface may crack and flake.
- **Overall:** Workability is poor to moderate; organic matter incorporation and careful irrigation improve tilth.

Nutrient Richness & Cation Exchange Capacity (CEC):

- **CEC:** Low to moderate (5–15 cmol(+)/kg), influenced by clay content.
- **Base saturation:** Very high (>70%) because of abundant sodium, calcium, and magnesium ions.
- **Nutrients:** Calcium and magnesium are adequate; nitrogen and phosphorus often deficient. High salinity reduces availability of potassium and micronutrients.
- **Organic matter:** Very low (<1%), leading to weak structure and limited biological activity.
- **Micronutrients:** Iron, zinc, and manganese deficiencies are common due to high pH and salt stress.

pH:

Typically alkaline (7.5–9.0). High pH combined with salinity reduces nutrient availability and may restrict crop growth.

Management:

- **Salinity management:** Apply gypsum or other calcium amendments to displace sodium; leach salts using adequate irrigation where drainage allows.
- **Organic matter:** Incorporate compost, green manure, or farmyard manure to improve structure and water retention.
- **Fertilization:** Apply nitrogen and phosphorus judiciously; use acid-forming fertilizers to counteract alkalinity.
- **Irrigation & drainage:** Ensure proper drainage to prevent salt accumulation; avoid over-irrigation that can raise the water table.
- **Erosion control:** Maintain vegetative cover and mulching to prevent wind erosion and crusting.

Suitable crops:

Drought- and salt-tolerant species such as barley, sorghum, millet, salt-tolerant legumes (e.g., cowpeas), and forage grasses. Under controlled irrigation and proper salinity management, some horticultural crops like tomatoes, onions, and melons may be cultivated.